Layer Sensitivity Matters: Mixed-Precision Post-Training Quantization for SAM2 Video Segmentation

Wenyu Zhou[1], **Jinyang Du**[2], **Yifu Ding**[2], **Chengtao Lv**[3], **Ying Li**[2*], **Xiaode Liu**[4,*] and **Jinyang Guo**[1]

[1]School of Artificial Intelligence, Beihang University, Beijing, China
[2]School of Computer Science and Engineering, Beihang University, Beijing, China
[3]Nanyang Technological University, Singapore
[4]Intelligent Science & Technology Academy of CASIC, China
wayaairon@gmail.com, jinyangdu@buaa.edu.cn, yifuding@buaa.edu.cn, chengtao001@e.ntu.edu.sg, liying@buaa.edu.cn, lxde@pku.edu.cn, jinyangguo@buaa.edu.cn

Abstract

Segment Anything Model 2 (SAM2) extends promptable segmentation to videos, but its streaming memory and transformer modules make deployment costly and make uniform low-bit post-training quantization (PTQ) fragile. We observe that SAM2's quantization sensitivity is highly non-uniform: a small set of memory-related and mask-decoding layers accounts for most degradation, while many other layers can remain low-bit. Based on this observation, we propose Greedy Layer-wise Sensitivity-guided Mixed-Precision Quantization (GLS-MPQ), a task-driven PTQ framework that starts from an all-low-bit SAM2 model and iteratively restores the layer with the largest marginal promptable video segmentation gain under an average-bit budget. To address the remaining video-memory bottleneck, we further combine GLS-MPQ with memory-aware KV compression, which merges redundant memory tokens and quantizes the resulting projected K/V sequence to 8 bits before memory attention. Experiments on DAVIS-2017 and SA-V with SAM2-B+ show that GLS-MPQ improves the accuracy-compression trade-off over uniform PTQ and PTQ4SAM-style baselines. With SAM2-B+ on DAVIS-2017, GLS-MPQ reaches 84.36 J&F under average 8-bit setting, only 0.60 below full precision, and M-KVM reduces end-to-end evaluation time by 17.4% while reaching 85.56 J&F. Additional results on SAM2-T indicate similar quantization sensitivity trends.

1 Introduction

Segment Anything Model (SAM) [Kirillov *et al.*, 2023] has shown strong generalization in promptable image segmentation, and SAM2 [Ravi *et al.*, 2025] further extends this capability to videos through a streaming memory mechanism. While effective for promptable video segmentation (PVS), SAM2 must repeatedly encode frames, maintain temporal memory, and decode object masks, hindering real-time deployment on resource-constrained devices.

Post-training quantization (PTQ) is an attractive compression technique because it avoids costly end-to-end retraining. Existing methods such as PTQ4SAM [Lv *et al.*, 2024] demonstrate the feasibility of PTQ for SAM-style models. However, SAM2 is not a frame-wise image model: its memory attention and propagation modules introduce additional numerical sensitivity. Consequently, uniform low-bit quantization can severely degrade video segmentation quality.

In this work, we revisit SAM2 PTQ from a layer-sensitivity perspective. We observe that uniform W6A6 causes substantial degradation on both DAVIS-2017 and SA-V, and W8A8 remains fragile on SA-V. In contrast, mixed low-bit/full-precision policies under parameter-weighted average 8-bit and 6-bit setting budgets recover much of the lost accuracy. This suggests that SAM2's quantization errors are highly non-uniform and concentrated in a small set of sensitive layers, rather than evenly distributed across the model.

Beyond numerical precision, SAM2 also suffers from a video-memory bottleneck. During propagation, current-frame queries repeatedly attend to cached keys and values from previous frames. Since the dominant memory-attention cost scales with the memory sequence length, reducing only K/V bit-width cannot fully address this bottleneck. This motivates joint compression of both the memory sequence length and the projected K/V representation.

We therefore propose Greedy Layer-wise Sensitivity-guided Mixed-Precision Quantization (GLS-MPQ) for SAM2. Starting from an all-low-bit model, GLS-MPQ iteratively restores the layer with the largest marginal PVS gain to full precision under a target average-bit budget. For average 8-bit setting, non-selected layers remain 6-bit; for average 6-bit setting, they remain 4-bit. To further reduce memory cost, we introduce Memory-Aware KV Merging (M-KVM), inspired by KVSlimmer [Liu *et al.*, 2026], which merges low-contribution adjacent memory KV pairs and quantizes the compressed projected K/V sequence to 8 bits before memory attention.

Our contributions are summarized as follows:

*Corresponding authors.

- We show that uniform low-bit PTQ is suboptimal for SAM2 video segmentation, since quantization degradation is highly non-uniform and concentrated in a few sensitive components.
- We propose GLS-MPQ, a greedy mixed-precision PTQ framework that restores layers with the largest downstream PVS marginal gains under an average-bit budget.
- We propose M-KVM, a memory-aware KV compression module that merges low-contribution projected memory tokens and quantizes the compressed K/V sequence to 8 bits before memory attention.
- Experiments on DAVIS [Perazzi *et al.*, 2016] and SA-V [Ravi *et al.*, 2025] show that GLS-MPQ improves the accuracy–compression trade-off; on SAM2-B+ DAVIS-2017, GLS-MPQ+M-KVM achieves 85.56 J&F with 17.4% lower end-to-end time.

2 Related Work

Efficient quantization. SAM2 extends SAM's promptable image segmentation to video through streaming memory and memory attention, making efficient deployment more demanding [Kirillov *et al.*, 2023; Ravi *et al.*, 2025]. PTQ compresses pretrained models without end-to-end retraining; PTQ4SAM adapts it to SAM, Q-MiniSAM2 establishes a SAM2 PTQ benchmark, and Q-SAM2 studies accurate low-bit SAM2 quantization [Lv *et al.*, 2024; Ren *et al.*, 2025; Farronato *et al.*, 2025]. These methods primarily improve calibration, reconstruction, or uniform low-bit settings rather than analyzing how quantization sensitivity varies across SAM2 layers. Prior mixed-precision methods use sensitivity estimates or automated search to assign per-layer precision [Wang *et al.*, 2019; Wu *et al.*, 2018]. Recent SAM2-oriented studies also explore mixed precision or low-bit quantization for efficient video segmentation [Duan *et al.*, 2026]. GLS-MPQ differs by using downstream PVS marginal gains and reevaluating candidates after each layer selection.

Memory token compression. Quantization does not shorten the memory sequence attended during SAM2 propagation. Q-MiniSAM2's Adaptive Mutual-KV reduces redundant cross-attention using inter-frame similarity [Ren *et al.*, 2025]. More broadly, visual-token methods prune input-dependent tokens or merge similar tokens to reduce transformer attention cost [Rao *et al.*, 2021; Bolya *et al.*, 2023], and KV-cache methods retain high-importance entries or merge redundant keys and values [Zhang *et al.*, 2023; Liu *et al.*, 2026]. These studies motivate sequence-length reduction, but SAM2 memory tokens encode spatial and temporal cues needed for mask propagation. M-KVM therefore merges low-contribution adjacent projected memory KV pairs and quantizes the compressed K/V sequence to 8 bits before memory attention.

3 Motivation

We motivate GLS-MPQ and M-KVM through two empirical observations.

Precision	DAVIS	Δ	SA-V	Δ
Full precision	84.96	–	72.42	–
Uniform W8A8	83.17	-1.79	56.47	-15.95
Uniform W6A6	78.72	-6.24	50.75	-21.67

Table 1: Uniform low-bit PTQ degrades SAM2-B+. Values are $\mathcal{J}\&\mathcal{F}$, and Δ is the drop from full precision.

3.1 Challenges in Uniform Quantization

Uniform quantization is efficient but can be fragile for architecture-sensitive models. Applying the same bit-width to all layers may over-compress sensitive components while wasting precision on robust ones. Our PTQ pipeline follows AdaRound [Nagel *et al.*, 2020] to optimize weight rounding from calibration data without end-to-end retraining. We first revisit the standard affine quantizer and then diagnose this issue in SAM2. For a tensor x, a uniform affine quantizer maps full-precision values to a low-bit representation through a scale s and zero point z:

$$\hat{x} = s\left(\mathrm{clip}\left(\lfloor x/s \rceil + z, q_{\min}, q_{\max}\right) - z\right). \tag{1}$$

The bit-width determines the integer range $[q_{\min}, q_{\max}]$, and WbAb denotes b-bit weight and activation quantization.

To localize the source of quantization degradation, we perform a hierarchical W4A4 sensitivity diagnosis over SAM2 components and submodules. As shown in Figure 1, the degradation is highly concentrated in a few components rather than uniformly distributed across the whole model. This motivates a bit-budgeted mixed-precision assignment that protects only the most fragile layers or modules.

We first apply uniform W8A8 and W6A6 PTQ to SAM2-B+ using the same PTQ4SAM-style [Lv *et al.*, 2024] calibration and reconstruction pipeline used in our main experiments. As shown in Table 1, W6A6 loses 6.24 points on DAVIS-2017 and 21.67 points on SA-V, while even W8A8 remains fragile on SA-V. Applying one precision to every target layer therefore imposes an unnecessarily strict constraint on robust layers while failing to protect the small set of layers that dominate the final PVS quality.

Our preliminary layer-restoration study further shows that retaining only seven selected layers in full precision, accounting for 5.72% of the model parameters, recovers much of this loss under the average 8-bit setting policy (Tables 2 and 3). This indicates that SAM2's quantization sensitivity is highly non-uniform and motivates a bit-budgeted mixed-precision assignment.

3.2 Redundancy in Video Memory

During PVS, current-frame queries Q repeatedly attend to cached memory keys and values (K, V). If Q contains n_q queries and the memory contains n_m tokens, the dominant attention cost scales as $O(n_q n_m d)$.

As illustrated in Figure 2, the memory-KV computation is dominated by the sequence-length-dependent attention term. In our SAM2-B+ DAVIS-2017 average 8-bit setting, the memory-KV portion contains 166.784G FLOPs before merging, where $QK^\top + AV$ contributes 165.725G FLOPs, accounting for 99.4% of this part. In contrast, the K/V projection contributes only 1.059G FLOPs. This indicates that the

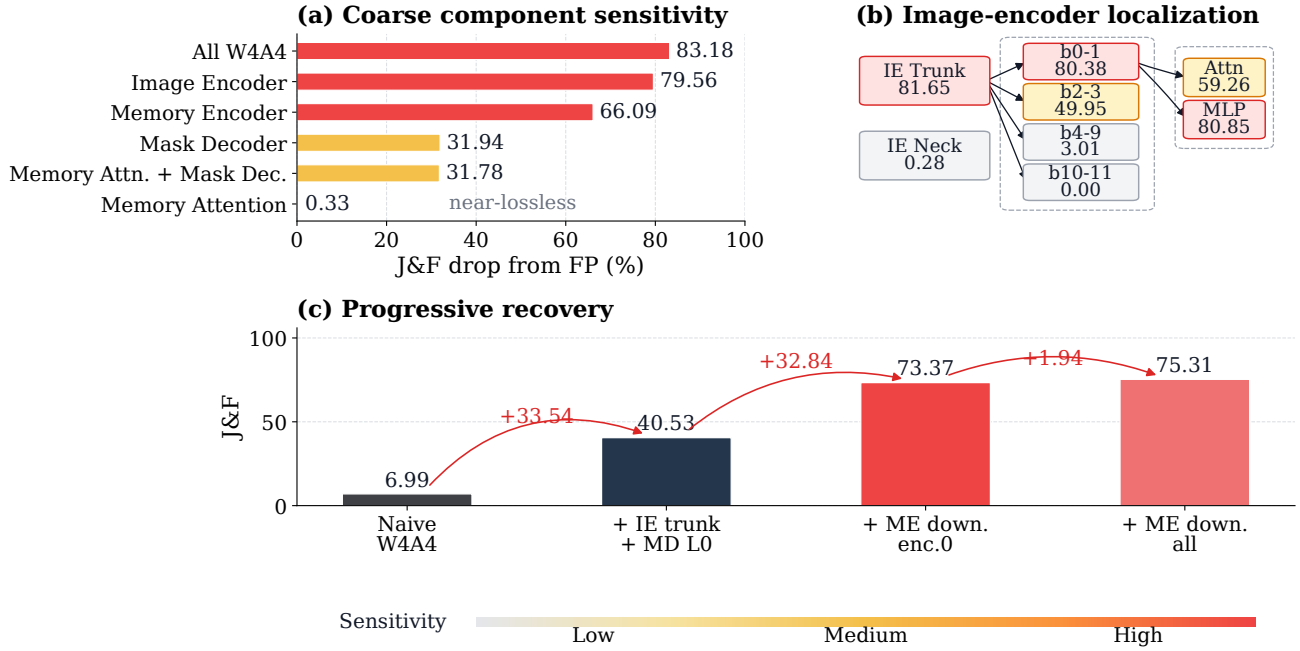


Figure 1: Hierarchical sensitivity diagnosis under W4A4 quantization on DAVIS-2017 with SAM2-B+. Quantization degradation is highly concentrated: image-encoder and memory-encoder quantization causes much larger degradation than memory attention alone. Progressive recovery shows that restoring a small set of sensitive modules can recover most of the lost accuracy. IE, ME, MD, Attn., Down., and Enc. denote image encoder, memory encoder, mask decoder, attention, downsampler, and encoder, respectively.

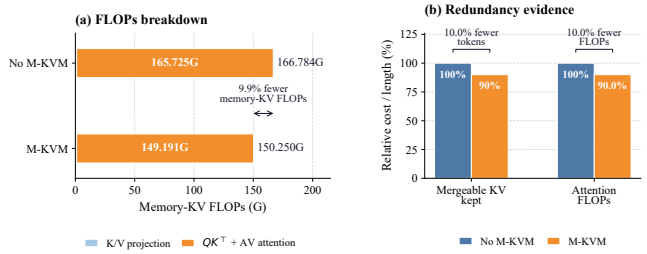


Figure 2: FLOPs evidence for redundancy in video memory. In the SAM2-B+ DAVIS-2017 average 8-bit setting, the $QK^\top + AV$ attention term accounts for 165.725G out of 166.784G memory-KV FLOPs before merging, showing that memory computation is dominated by the sequence-length-dependent attention term. Keeping 90% of mergeable memory K/V tokens reduces this attention term to 149.191G and the memory-KV FLOPs to 150.250G.

main computational redundancy lies in repeatedly attending to a long memory sequence, rather than in the projection itself. Therefore, reducing only the bit-width of K/V cannot address the dominant FLOPs term; shortening the attended memory sequence is also necessary.

Quantization reduces numerical representation cost but does not reduce n_m. A naive token drop or a merge based only on static similarity is risky because the usefulness of a memory token depends on the current query, and SAM2 also contains special memory suffix tokens that must be preserved.

As a preliminary memory-side compression observation, keeping 90% of the mergeable projected memory K/V tokens and quantizing the resulting K/V sequence to 8 bits shortens the end-to-end evaluation time from 2h 24min to 1h 59min in the SAM2-B+ DAVIS-2017 average 8-bit setting. This suggests useful memory redundancy, but safe compression must preserve query-relevant information rather than remove tokens indiscriminately. It motivates Memory-Aware KV Merging (M-KVM), which selects low-contribution adjacent pairs using the current memory attention and applies 8-bit quantization after merging.

4 Method

Figure 3 gives an overview of the proposed framework, which combines sensitivity-guided mixed-precision quantization for SAM2 weights with memory-aware KV merging and 8-bit projected K/V quantization.

4.1 Mixed-Precision Quantization Framework

We introduce the mixed-precision quantization pipeline that converts a calibrated low-bit SAM2 model into a layer-wise precision assignment under a fixed storage budget. Let f_θ denote a pretrained SAM2 model and let $\mathcal{D}_{cal}$ be a calibration set. Instead of using one bit-width for all L target layers, we search for a layer-wise assignment $q = \{q_1, q_2, \ldots, q_L\}$, where each q_l is selected from a candidate precision set $\mathcal{P}$ such as $\{4\text{-bit}, 6\text{-bit}, \mathrm{FP}\}$. Let $\mathcal{M}(q)$ denote the PVS metric of an assignment and let $\mathcal{C}(q)$ denote its parameter storage cost. The mixed-precision objective is

$$\min_q \mathcal{C}(q) \quad \text{s.t.} \quad \mathcal{M}(q) \geq \mathcal{M}_{\mathrm{FP}} - \epsilon, \tag{2}$$

where ϵ is the allowed performance drop.

We use an effective average weight bit-width to express the

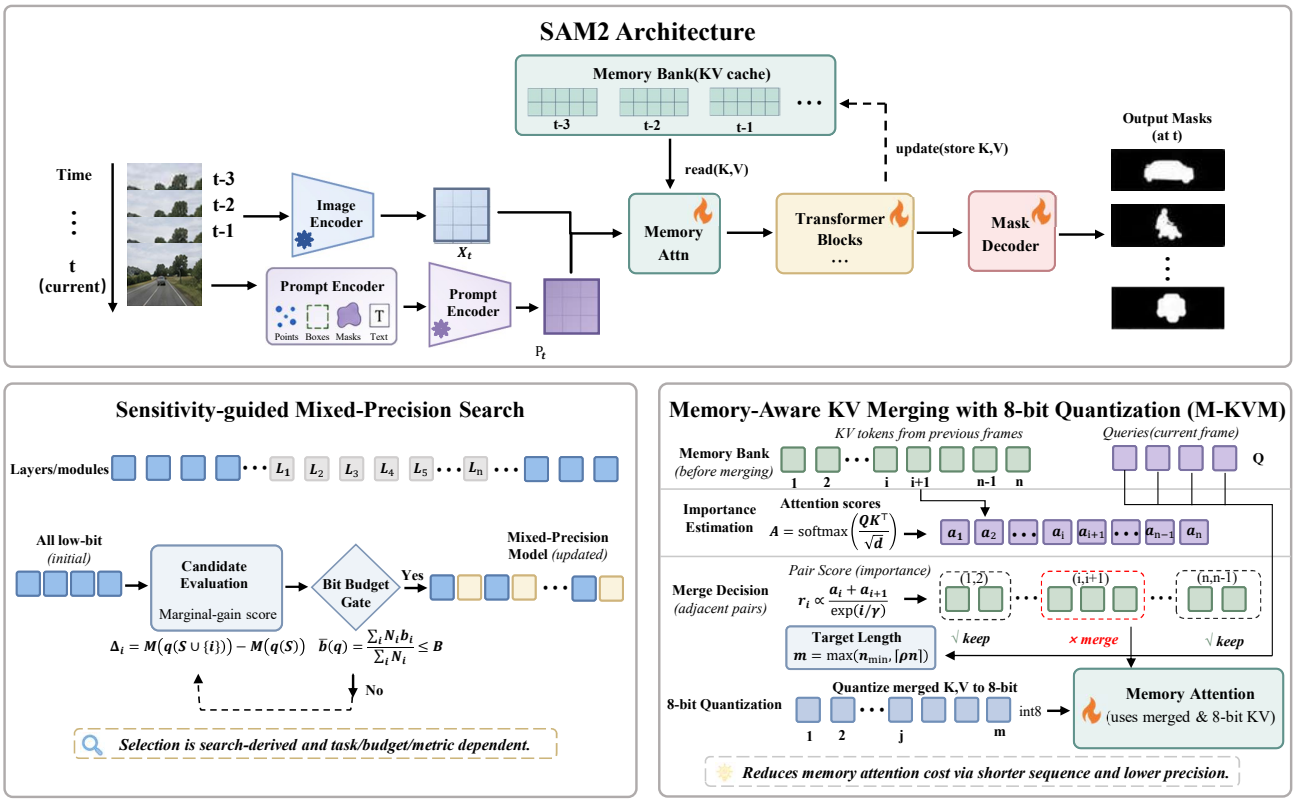


Figure 3: Overview of the proposed GLS-MPQ and M-KVM framework. GLS-MPQ searches for a mixed-precision assignment by restoring layers with the largest task-level marginal gain under an average-bit budget. M-KVM reduces memory-attention cost by query-conditioned adjacent KV merging and 8-bit projected K/V quantization.

storage budget:

$$\bar{b}(q) = \frac{\sum_{l=1}^{L} N_l b_l(q_l)}{\sum_{l=1}^{L} N_l}, \tag{3}$$

where N_l is the number of parameters in layer l and $b_l(q_l)$ is its assigned weight bit-width. Here, q_l denotes the precision choice of layer l, while $b_l(q_l)$ gives its numerical bit-width. The sum covers all weight-bearing modules used in model-size accounting, including modules that are always kept in full precision. Full-precision layers are counted as FP32, i.e., $b_l = 32$, because our evaluation code loads SAM2 weights in float32 without FP16/BF16 autocasting. We use $\bar{b}(q) \leq B + \tau$ with $B \in \{6, 8\}$, where τ is a tolerance for whole-layer precision granularity and is set to 0.1 in our reported models.

The key challenge is the cost of finding q. Exhaustive assignment requires $|\mathcal{P}|^L$ task evaluations. Even if the search starts from one low-bit model and only restores at most $K_{\max}$ layers to full precision, a naive combination search requires $\sum_{k=0}^{K_{\max}} \binom{L}{k}$ evaluations. Each evaluation runs PVS on validation videos, making this search impractical. A one-shot sensitivity ranking reduces the cost to $O(L)$ but measures every layer in the same all-low-bit context and ignores interactions after layers are restored. We therefore use a greedy search that recomputes task-level marginal gains after each selection.

4.2 Greedy Layer-Wise Sensitivity Search

Motivation. The mixed-precision objective defines a layer-wise bit-width assignment under a storage budget, but solving it directly is prohibitively expensive because the search space grows combinatorially with the number of target layers. A one-shot sensitivity ranking is cheaper, yet it evaluates each layer in a fixed all-low-bit context and cannot reflect how layer importance changes after previous layers have been restored. We therefore adopt a greedy layer-wise search that starts from an all-low-bit model and repeatedly restores the feasible layer with the largest validation gain.

Selection rule. For a target low-bit precision $p_{\text{low}} \in \{4\text{-bit}, 6\text{-bit}\}$ and a high precision $p_{\text{high}} = \text{FP}$, we first construct an all-quantized baseline $q^{(0)} = \{p_{\text{low}}\}_{l=1}^{L}$ and evaluate $M_{\text{base}} = \mathcal{M}(q^{(0)})$. This baseline is the starting point of the search. Let $\mathcal{S}$ denote the set of layers already restored to full precision and let $\mathcal{U}$ be the remaining candidate layers. We use $q(\mathcal{S})$ to denote the precision assignment that keeps layers in $\mathcal{S}$ at p_{high} and all other target layers at p_{low}. At iteration k, we first form the feasible candidate set

$$\mathcal{F}^{(k)} = \{i \in \mathcal{U} \mid \bar{b}(q(\mathcal{S} \cup \{i\})) \leq B + \tau\}. \tag{4}$$

For each feasible candidate layer $i \in \mathcal{F}^{(k)}$, we evaluate the marginal gain:

$$\Delta_i^{(k)} = \mathcal{M}(q(\mathcal{S} \cup \{i\})) - \mathcal{M}(q(\mathcal{S})). \tag{5}$$

Algorithm 1 GLS-MPQ for SAM2

Input: Full-precision SAM2 model f_θ, calibration data $\mathcal{D}_{cal}$, validation data $\mathcal{D}_{val}$, low-bit precision p_{low}, target average bit budget B, tolerance τ, optional FP layer cap $K_{\max}$, target metric M_{target}, gain threshold δ.
Output: Mixed-precision quantized model $f_{\theta,q}$.

Calibrate quantizers on $\mathcal{D}_{cal}$.
Quantize all target layers to p_{low}.
Evaluate the all-quantized metric M_{base}.
Initialize $\mathcal{S} \leftarrow \emptyset$ and $\mathcal{U} \leftarrow \{1, \dots, L\}$.
while $\bar{b}(q(\mathcal{S})) < B + \tau$, $|\mathcal{S}| < K_{\max}$ and $\mathcal{M}(q(\mathcal{S})) < M_{\text{target}}$ **do**
  $\mathcal{F} \leftarrow \{i \in \mathcal{U} \mid \bar{b}(q(\mathcal{S} \cup \{i\})) \le B + \tau\}$.
  if $\mathcal{F} = \emptyset$ **then**
    break
  end if
  for each candidate layer $i \in \mathcal{F}$ **do**
    Restore layers $\mathcal{S} \cup \{i\}$ to FP and keep $\mathcal{U} \setminus \{i\}$ at p_{low}.
    Compute the marginal gain Δ_i on $\mathcal{D}_{val}$.
  end for
  Select $i^* = \arg\max_{i \in \mathcal{F}} \Delta_i$.
  if $\Delta_{i^*} < \delta$ **then**
    break
  end if
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{i^*\}, \mathcal{U} \leftarrow \mathcal{U} \setminus \{i^*\}$.
end while
Keep layers in $\mathcal{S}$ in FP and all other target layers at p_{low}.
return the mixed-precision model $f_{\theta,q}$.

The feasible layer with the largest marginal gain is selected:

$$i^{(k)} = \arg\max_{i \in \mathcal{F}^{(k)}} \Delta_i^{(k)}, \quad \mathcal{S} \leftarrow \mathcal{S} \cup \{i^{(k)}\}. \tag{6}$$

Unlike one-shot sensitivity ranking, GLS-MPQ recomputes marginal gains after each restoration step, allowing the search to account for interactions among SAM2 layers.

Stopping Rule and Precision Assignment. The greedy process stops when $\mathcal{M}(q(\mathcal{S})) \ge M_{\text{target}}$, $\mathcal{F}^{(k)} = \emptyset$, or $\max_{i \in \mathcal{F}^{(k)}} \Delta_i^{(k)} < \delta$. Here $M_{\text{target}} = \mathcal{M}_{\text{FP}} - \epsilon$ is the target accuracy and δ is a minimum useful gain threshold. We optionally keep $K_{\max}$ as a secondary cap on the number of FP layers, but the primary constraint is the target average bit budget B with tolerance τ. The final assignment keeps $q_l = p_{\text{high}}$ for $l \in \mathcal{S}$ and $q_l = p_{\text{low}}$ otherwise. We name the resulting policies by their effective average bit budget rather than by p_{low}: average 8-bit setting uses 6-bit non-selected layers plus selected FP layers under $B = 8$, while average 6-bit setting uses 4-bit non-selected layers plus selected FP layers under $B = 6$. With at most $K_{\max}$ restoration steps, GLS-MPQ requires at most $O(K_{\max} L)$ task evaluations rather than the combinatorial search described above.

4.3 Memory-Aware KV Merging and Quantization

Motivated by the memory redundancy observed in Figure 2, we propose Memory-Aware KV Merging (M-KVM), which shortens and quantizes the projected memory KV sequence immediately before memory attention while keeping the GLS-MPQ layer assignment unchanged. A direct token drop can discard useful values, and a static similarity score does not reflect relevance to the current query. Inspired by the adjacent KV merging principle of KVSlimmer [Liu *et al.*, 2026], M-KVM instead uses query-conditioned attention mass to select low-contribution adjacent pairs, merges rather than deletes their information, and protects SAM2's special suffix tokens.

Let the projected and head-split memory tensors be $K = [K_m; K_e]$ and $V = [V_m; V_e]$, where K_m, V_m are the mergeable memory tokens and K_e, V_e are the excluded suffix tokens, such as SAM2 object pointer tokens that are also excluded from RoPE. If $n = |K_m|$ and the keep ratio is ρ, M-KVM sets the target mergeable length to $m = \max(n_{\min}, \lceil \rho n \rceil)$, where $n_{\min}$ prevents degenerate caches. Thus the threshold is a token budget rather than a learned similarity cutoff: M-KVM removes $n - m$ mergeable tokens and keeps the excluded suffix length unchanged.

To decide which KV pairs to merge, M-KVM first computes the attention mass of each mergeable key,

$$A = \text{softmax}\left(Q[K_m; K_e]^\top / \sqrt{d}\right), \quad a_{b,h,i} = \sum_t A_{b,h,t,i}, \tag{7}$$

where $a_{b,h,i}$ is the attention mass of token i for batch element b and head h. For each adjacent pair $(i, i+1)$, we use the position-decayed pair score

$$r_i = \frac{1}{BH} \sum_{b,h} \frac{a_{b,h,i} + a_{b,h,i+1}}{\exp(i/\gamma)}. \tag{8}$$

We set $\gamma = 4096/\log 512$ in the implementation. The non-overlapping adjacent pairs with the smallest r_i are merged first, since they contribute least to the current memory attention. In each round, we select up to the number of pairs needed to reach the target length m and then recompute attention mass if further merging is required.

For a selected pair $(i, i+1)$, the value vectors are merged by addition, $\tilde{v}_i = v_i + v_{i+1}$. The key vectors use a Hessian-proxy closed-form merge inspired by KVSlimmer. Omitting (b, h) for readability, with the per-head normalized mass $\alpha_i = a_i / (\sum_{j=1}^{n} a_j + \varepsilon)$, $o = \sum_j \alpha_j v_j$, and $d_i = \|v_i - o\|_1$, we compute

$$h_{ii} = \alpha_i (1 - 2\alpha_i) d_i, \quad h_{i,i+1} = \alpha_i \alpha_{i+1} (d_i + d_{i+1}). \tag{9}$$

The merged key is

$$\tilde{k}_i = \frac{(h_{ii} - h_{i,i+1}) k_i + (h_{i+1,i+1} - h_{i,i+1}) k_{i+1}}{h_{ii} + h_{i+1,i+1} - 2h_{i,i+1} + \varepsilon}. \tag{10}$$

Let the resulting compressed mergeable sequences be $\tilde{K}_m, \tilde{V}_m$. Unselected tokens are copied to these sequences, and K_e, V_e are concatenated back without merging. We then apply an 8-bit asymmetric affine quantizer Q_8 to the complete projected K/V sequence:

$$\hat{K} = Q_8([\tilde{K}_m; K_e]), \qquad \hat{V} = Q_8([\tilde{V}_m; V_e]), \tag{11}$$

and use $(\hat{K}, \hat{V})$ in memory attention. Quantization reduces the representation cost of each retained token, while merging

reduces the sequence-length cost. In our experiments, we use $\rho = 0.90$, $n_{\min} = 1$, and 8-bit projected K/V quantization, and report runtime for the average 8-bit setting.

5 Experiments

5.1 Experimental Setup

We evaluate SAM2.1 Hiera-T and SAM2.1 Hiera-B+ to test whether the observed sensitivity pattern is consistent across compact and larger SAM2 variants.

Datasets and task. Experiments are conducted on DAVIS-2017 and SA-V under the PVS setting. DAVIS is evaluated on all 30 videos with an annotation stride of 1, and SA-V is evaluated on all 155 videos with annotation stride 4. Full-precision and quantized models use the same offline interactive prompting protocol: each object starts from three positive oracle clicks on the first valid frame, followed by at most eight refinement passes that add three clicks on the worst unprompted frame.

Metrics. The main metric is the standard region-and-boundary score $\mathcal{J}\&\mathcal{F}$. We also report $\mathcal{J}$ and $\mathcal{F}$, averaged over videos. Runtime is measured end to end, including prompt propagation and mask evaluation.

Quantization settings. Unless otherwise specified, we quantize target linear and convolutional layers in the image encoder, memory attention, memory encoder, and mask decoder. We keep normalization layers, embeddings, the patch embedding, mask-decoder output upscaling layers, the IoU prediction head, and output hypernetwork MLPs in full precision. For the Q-MiniSAM2/AdaptiveMutualKV comparison, the memory encoder is additionally kept in full precision.

For uniform W8A8/W6A6 baselines and the low-bit layers used by GLS-MPQ, we follow AdaRound [Nagel *et al.*, 2020] for weight quantization and use asymmetric activation quantization. For all M-KVM results, the merged projected memory K/V sequence is additionally quantized to 8 bits with asymmetric affine quantization.

Calibration and reconstruction. For standard SAM2 PTQ, we use 32 SA-V videos with annotation stride 4 for calibration and the first five calibration videos for reconstruction. Following Q-MiniSAM2 [Ren *et al.*, 2025], we also include a comparison setting with 32 SA-V videos, at most 20 annotated frames per video, and HVQ triplets from the first five calibration videos. Ground-truth masks are used only to generate calibration prompts, not as reconstruction losses.

Baselines. We compare GLS-MPQ with full-precision SAM2, uniform low-bit PTQ, AdaRound [Nagel *et al.*, 2020], QDrop [Wei *et al.*, 2022], PTQ4SAM [Lv *et al.*, 2024], and Q-MiniSAM2 [Ren *et al.*, 2025]. We also report GLS-MPQ combined with the proposed M-KVM module. We report both W8A8 and W6A6 settings when available. For GLS-MPQ, greedy-selected layers are used as full-precision skip layers, while all non-selected target layers use 6-bit in average 8-bit setting and 4-bit in average 6-bit setting. Thus average 8-bit setting and average 6-bit setting denote parameter-weighted average bit budgets, not uniform bit-widths. Full-precision parameters are counted as FP32 when computing the effective average bit-width.

Method	FP	8-bit budget	6-bit budget
DAVIS-2017, SAM2-B+			
AdaRound	84.96	62.74	38.06
QDrop		82.16	80.13
PTQ4SAM		83.17	78.72
Q-MiniSAM2		83.26	80.55
GLS-MPQ		84.36	**80.68**
GLS-MPQ+M-KVM		**85.56**	79.94
DAVIS-2017, SAM2-T			
AdaRound	83.78	77.53	**76.66**
QDrop		78.36	73.65
PTQ4SAM		77.26	73.97
Q-MiniSAM2		77.17	73.74
GLS-MPQ		**80.70**	75.90
GLS-MPQ+M-KVM		80.01	74.93
SA-V, SAM2-B+			
AdaRound	72.42	51.04	42.27
QDrop		55.32	45.31
PTQ4SAM		56.47	50.75
Q-MiniSAM2		61.29	50.09
GLS-MPQ		69.56	**56.82**
GLS-MPQ+M-KVM		**69.83**	53.18

Table 2: Main accuracy results. The FP column reports the full-precision reference accuracy for each dataset–model setting. For fixed-bit baselines, the 8-bit and 6-bit budget columns correspond to W8A8 and W6A6. For GLS-MPQ, they correspond to mixed-precision average 8-bit and 6-bit setting under matched effective average bit budgets. M-KVM denotes Memory-Aware KV Merging with a keep ratio $\rho = 0.90$ and 8-bit projected memory K/V quantization.

5.2 Main Accuracy Results

Table 2 summarizes the main accuracy results on DAVIS-2017 and SA-V. On DAVIS-2017 with SAM2-B+, GLS-MPQ reaches 84.36 under average 8-bit setting, only 0.60 points below full precision and 1.10 points higher than the strongest fixed-bit baseline. Under average 6-bit setting, GLS-MPQ reaches 80.68 and remains 0.13 points above the strongest fixed-bit baseline. On SA-V with SAM2-B+, GLS-MPQ improves over the strongest fixed-bit baseline by 8.27 and 6.07 points under average 8-bit and 6-bit setting, respectively. These results show that sensitivity-guided mixed precision is especially beneficial when the uniform low-bit baseline is fragile. Table 3 reports the actual parameter-weighted average bit-widths of the two GLS-MPQ policies. A plausible explanation for the large uniform-PTQ degradation is that video memory propagates small numerical errors across frames while mask decoding requires precise spatial predictions. GLS-MPQ mitigates this non-uniform sensitivity by spending full precision only on the layers that matter most. The average 8/6-bit setting gap also exposes a deployment trade-off: average 8-bit setting nearly preserves the full-precision DAVIS accuracy, whereas average 6-bit setting provides stronger storage compression at a larger accuracy cost.

Policy	Low-bit	B	FP layers	Selected FP	Total FP	$\bar{b}$
Avg-8	W6A6	8	7	5.72%	7.44%	7.94
Avg-6	W4A4	6	7	5.72%	7.44%	6.08

Table 3: Effective average bit budgets for the SAM2-B+ GLS-MPQ models. B is the target budget; Selected FP and Total FP report parameter shares for the greedy-selected layers and all full-precision modules, respectively. FP is counted as FP32 in $\bar{b}$. The average 6-bit setting policy slightly exceeds 6.0 because whole layers are restored under the $\tau = 0.1$ budget tolerance.

5.3 Greedy Search Analysis

We analyze the greedy search by recording the selected FP layers and their marginal gains $\Delta_i^{(k)}$. The results show highly non-uniform gains, confirming that only a small subset of layers needs FP protection. Since each gain is recomputed after previous selections, the selected order can differ from a one-shot sensitivity ranking and better reflects the current mixed-precision state. The selected layers may still depend on the validation subset, prompt protocol, and evaluation dataset.

5.4 Ablation Study

Table 2 also isolates the effect of sensitivity-guided precision assignment on SA-V with SAM2-B+. Compared with the strongest fixed-bit baselines, GLS-MPQ improves $\mathcal{J}\&\mathcal{F}$ by 8.27 and 6.07 points under average 8-bit and 6-bit setting, respectively. This confirms that restoring sensitive layers is more effective than applying one low-bit precision uniformly.

To test whether the improvement can be obtained by restoring arbitrary layers, we compare GLS-MPQ with five random two-layer FP selections on DAVIS-2017 using SAM2-T. GLS-MPQ reaches 80.70 and 75.90 under average 8-bit and 6-bit setting, whereas the random selections average 75.39 and 60.99, respectively. The complete random trials and selected layers are reported in Appendix A.

Table 2 ablates M-KVM by comparing GLS-MPQ with and without its memory merging and 8-bit projected K/V quantization. Under average 8-bit setting, M-KVM changes $\mathcal{J}\&\mathcal{F}$ by $+1.20$, -0.69, and $+0.27$ points on DAVIS SAM2-B+, DAVIS SAM2-T, and SA-V SAM2-B+, respectively. Under average 6-bit setting, the corresponding changes are -0.74, -0.97, and -3.64 points. These results show that M-KVM can improve the average 8-bit SAM2-B+ setting, but its effect is model- and precision-dependent and becomes less reliable under the more aggressive average 6-bit setting policy. We therefore treat M-KVM as a memory-compression module rather than the core model-weight quantization contribution.

5.5 Efficiency Analysis

Figure 4 summarizes the practical effect of M-KVM. On DAVIS-2017 with SAM2-B+ under average 8-bit setting, M-KVM improves J&F from 84.36 to 85.56 and reduces end-to-end evaluation time from 144 minutes to 119 minutes.

GLS-MPQ and M-KVM reduce different deployment costs. GLS-MPQ reduces weight storage through its mixed-precision assignment. Because quantization does not change operator shapes, its conventional FLOP count is unchanged;

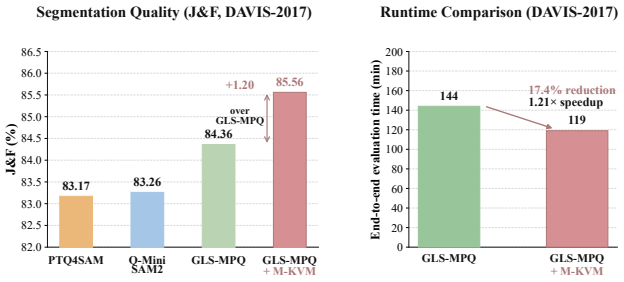


Figure 4: Accuracy and runtime comparison on DAVIS-2017 with SAM2-B+ under average 8-bit setting. M-KVM improves J&F from 84.36 to 85.56 and reduces end-to-end evaluation time from 144 minutes to 119 minutes, corresponding to a 17.4% reduction and a 1.21× speedup.

GLS-MPQ weight-storage estimate Policy	$\bar{b}$	Storage vs. FP32	Compression
FP32	32.00	100.0%	1.00×
Avg-8	7.94	24.8%	4.03×
Avg-6	6.08	19.0%	5.26×
M-KVM projected memory-KV efficiency Metric	No M-KVM	M-KVM	Reduction
Mergeable KV kept	100.0%	90.0%	10.0%
KV bit-width	32	8	75.0%
Est. KV storage	100.0%	22.5%	77.5%
K/V projection FLOPs	1.059 G	1.059 G	0.0%
$QK^\top + AV$ FLOPs	165.725 G	149.191 G	10.0%
End-to-end time	2h 24min	1h 59min	17.4%

Table 4: Deployment efficiency on SAM2-B+. GLS-MPQ storage excludes quantization metadata. M-KVM is measured on DAVIS-2017 with GLS-MPQ Avg-8, $\rho = 0.90$, and 8-bit projected K/V quantization; KV storage covers mergeable tokens only.

we therefore report the directly estimable weight-storage cost. Using the effective average weight bit-widths in Table 3, and counting full-precision parameters as FP32, the estimated weight-storage ratio relative to FP32 is $\bar{b}/32$. Average 8-bit and 6-bit setting therefore require 24.8% and 19.0% of the FP32 weight storage, corresponding to $4.03\times$ and $5.26\times$ compression. These estimates exclude small quantization metadata, and practical inference speedup depends on hardware support for mixed low-bit computation.

M-KVM reduces both the sequence length and representation cost of projected memory K/V. For the mergeable tokens, retaining 90% and using 8-bit values requires an estimated $0.90 \times 8/32 = 22.5\%$ of the uncompressed FP32 KV storage, corresponding to a 77.5% reduction or $4.44\times$ compression. The 8-bit quantization does not change conventional FLOPs. As shown in Table 4, merging occurs after K/V projection and therefore leaves its FLOPs unchanged, while the $QK^\top + AV$ attention FLOPs decrease from 165.725 G to 149.191 G. Across projection and attention, the reported memory-KV FLOPs decrease from 166.784 G to 150.250 G, or 9.9%. The end-to-end DAVIS-2017 evaluation time decreases from 2h 24min to 1h 59min, a 17.4% reduction, while improving average 8-bit setting's accuracy by 1.20 points in

this setting.

6 Conclusion

We studied post-training quantization for SAM2 under the PVS setting. Direct uniform low-bit quantization causes noticeable degradation, but layer-wise sensitivity analysis reveals that the degradation is concentrated in a small set of fragile components. Based on this finding, we proposed GLS-MPQ, a greedy task-driven mixed-precision PTQ strategy that starts from an all-low-bit model and iteratively restores the layers with the largest marginal gains to full precision. Experiments on DAVIS and SA-V with SAM2-T and SAM2-B+ show that the average 8-bit and 6-bit setting average-bit policies provide a better accuracy-efficiency trade-off than uniform low-bit quantization. M-KVM further shows that jointly reducing memory-token count and projected K/V precision can reduce runtime and memory cost on top of GLS-MPQ.

Acknowledgements

This work was supported by the Fundamental Research Funds for the Central Universities, and the National Natural Science Foundation of China (Nos. 62306025, 92367204).

Appendix

A Random Layer Selection

Table 5 reports five random two-layer FP selections under the same SAM2-T evaluation setting as the greedy policy. Although one random draw approaches GLS-MPQ, random selection is unstable and is lower by 5.31 and 14.91 points on average under average 8-bit and 6-bit setting, respectively.

Selection	Avg-8	Avg-6
GLS-MPQ (greedy)	**80.70**	**75.90**
Random 1	73.29	55.79
Random 2	74.73	59.26
Random 3	80.56	75.19
Random 4	72.74	56.47
Random 5	75.63	58.22
Random mean	75.39	60.99

Table 5: Greedy versus random two-layer FP selection on DAVIS-2017 using SAM2-T. Values are $\mathcal{J}\&\mathcal{F}$.

Random layer sets. Using shortened module names, Random 1 restores memory-encoder downsampler layer 0 and mask-decoder transformer layer 1; Random 2 restores downsampler layer 1 and mask-decoder layer 1; Random 3 restores downsampler layer 1 and mask-decoder layer 0; Random 4 restores image-encoder block 0 and memory-attention layer 0; and Random 5 restores image-encoder block 13 and mask-decoder layer 1.

References

[Bolya *et al.*, 2023] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your ViT but faster. In *International Conference on Learning Representations*, 2023.

[Duan *et al.*, 2026] Yuzhe Duan, Xuanxuan Ren, Guizhe Dong, Xu Yang, and Yanhua Yang. Mix-QSAM2: Mixed-precision quantization for high fidelity segmentation in resource constrained scenarios. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 3741–3749, 2026.

[Farronato *et al.*, 2025] Nicola Farronato, Florian Scheidegger, Mattia Rigotti, Cristiano Malossi, Michele Magno, and Haotong Qin. Q-SAM2: Accurate quantization for segment anything model 2. arXiv preprint arXiv:2506.09782, 2025.

[Kirillov *et al.*, 2023] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023.

[Liu *et al.*, 2026] Lianjun Liu, Hongli An, Weiqi Yan, Xin Du, Shengchuan Zhang, Huazhong Liu, and Yunshan Zhong. KVSlimmer: Theoretical insights and practical

optimizations for asymmetric KV merging. arXiv preprint arXiv:2603.00907, 2026.

[Lv *et al.*, 2024] Chengtao Lv, Hong Chen, Jinyang Guo, Yifu Ding, and Xianglong Liu. PTQ4SAM: Post-training quantization for segment anything. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15941–15951, 2024.

[Nagel *et al.*, 2020] Markus Nagel, Rana Ali Amjad, Mart van Baalen, Christos Louizos, and Tijmen Blankevoort. Up or down? adaptive rounding for post-training quantization. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7197–7206. PMLR, 2020.

[Perazzi *et al.*, 2016] Federico Perazzi, Jordi Pont-Tuset, Brian McWilliams, Luc Van Gool, Markus Gross, and Alexander Sorkine-Hornung. A benchmark dataset and evaluation methodology for video object segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 724–732, 2016.

[Rao *et al.*, 2021] Yongming Rao, Wenliang Zhao, Benlin Liu, Jiwen Lu, Jie Zhou, and Cho-Jui Hsieh. DynamicViT: Efficient vision transformers with dynamic token sparsification. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[Ravi *et al.*, 2025] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, 2025.

[Ren *et al.*, 2025] Xuanxuan Ren, Xiangyu Li, Kun Wei, Xu Yang, and Yanhua Yang. Q-MiniSAM2: A quantization-based benchmark for resource-efficient video segmentation. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 1829–1837. International Joint Conferences on Artificial Intelligence Organization, 2025.

[Wang *et al.*, 2019] Kuan Wang, Zhijian Liu, Yujun Lin, Ji Lin, and Song Han. HAQ: Hardware-aware automated quantization with mixed precision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8612–8620, 2019.

[Wei *et al.*, 2022] Xiuying Wei, Ruihao Gong, Yuhang Li, Xianglong Liu, and Fengwei Yu. QDrop: Randomly dropping quantization for extremely low-bit post-training quantization. In *International Conference on Learning Representations*, 2022.

[Wu *et al.*, 2018] Bichen Wu, Yanghan Wang, Peizhao Zhang, Yuandong Tian, Peter Vajda, and Kurt Keutzer. Mixed precision quantization of ConvNets via differentiable neural architecture search. arXiv preprint arXiv:1812.00090, 2018.

[Zhang *et al.*, 2023] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.